1. Find the smallest number in an array
2. Find the largest number in an array
3. Second Smallest and Second Largest element in an array
4. Reverse a given array
5. Count frequency of each element in an array
6. Rearrange array in increasing-decreasing order
7. Calculate sum of the elements of the array
8. Rotate array by K elements – Block Swap Algorithm
9. Average of all elements in an array
10. Find the median of the given array
11. Remove duplicates from a sorted array
12. Remove duplicates from unsorted array
13. Adding Element in an array
14. Find all repeating elements in an array
15. Find all non-repeating elements in an array
16. Find all symmetric pairs in array
17. Maximum product subarray in an array
18. Replace each element of the array by its rank in the array
19. Sorting elements of an array by frequency
20. Rotation of elements of array- left and right
21. Finding equilibrium index of an array
22. Finding Circular rotation of an array by K positions
23. Sort an array according to the order defined by another array
24. Search an element in an array
25. Check if Array is a subset of another array or notProblems on Numbers

Number-based problems are fundamental in competitive programming. These will help you improve your mathematical reasoning and logic:

26.
27. Check if a number is palindrome or not
28. Find all Palindrome numbers in a given range
29. Check if a number is prime or not

30. Prime numbers in a given range
31. Check if a number is armstrong number of not
32. Check if a number is perfect number
33. Even or Odd
34. Check weather a given number is positive or negative
35. Sum of first N natural numbers
36. Find Sum of AP Series
37. Program to find sum of GP Series
38. Greatest of two numbers
39. Greatest of three numbers
40. Leap Year or not
41. Reverse digits of a number
42. Maximum and Minimum digit in a number
43. Print Fibonacci upto Nth Term
44. Factorial of a number
45. Power of a number
46. Factors of a given number
47. Print all prime factors of the given number
48. Check if a number is a strong number or not
49. Check if a Number is Automorphic
50. GCD of two numbers
51. LCM of two numbers
52. Check if a number is Harshad number
53. Check if the number is abundant number or not
54. Sum of digits of a number
55. Sum of numbers in the given range
56. Permutations in which N people can occupy R seats in a classroom
57. Program to add two fractions
58. Replace all 0s with 1s in a given integer
59. Can a number be expressed as a sum of two prime numbers
60. Calculate the area of circle

61. Program to find roots of a Quadratic EquationProblems on Number System

These problems will test your understanding of number systems, a crucial concept for many coding interviews and exams:

62.
63. Convert Binary to Decimal
64. Convert binary to octal
65. Decimal to Binary conversion
66. Convert decimal to octal
67. Convert octal to binary
68. Convert octal to decimal
69. Convert digits/numbers to wordsProblems on Sorting

Sorting algorithms are essential, and understanding the different techniques will give you an edge in problem-solving:

70.
71. Bubble Sort Algorithm
72. Selection Sort Algorithm
73. Insertion Sort Algorithm
74. Quick Sort Algorithm
75. Merge sort algorithm

Problems on Strings

String manipulation is an important skill, and there are many practical problems that require efficient string handling:

76.
77. Check if a given string is palindrome or not
78. Count number of vowels, consonants, spaces in String
79. Find the ASCII value of a character
80. Remove all vowels from the string
81. Remove spaces from a string

82. Remove characters from a string except alphabets
83. Reverse a String
84. Remove brackets from an algebraic expression
85. Sum of the numbers in a String
86. Capitalize first and last character of each word
87. Calculate frequency of characters in a string
88. Find Non-repeating characters of a String
89. Check if two strings are anagram of each other
90. Count common sub-sequence in two strings
91. Check if two strings match where one string contains wildcard characters
92. Return maximum occurring character in the input string
93. Remove all duplicates from the input string.
94. Print all the duplicates in the input string.
95. Remove characters from first string present in the second string
96. Change every letter with the next lexicographic alphabet in the given string
97. Write a program to find the largest word in a given string.
98. Write a program to sort characters in a string
99. Count number of words in a given string
100. Write a program to find a word in a given string which has the highest number of repeated letters
101. Change case of each character in a string
102. Concatenate one string to another
103. Write a program to find a substring within a string. If found display its starting position
104. Reverse words in a string